

Packet Tracer Practice Lab 02

Part 1: IP Calculation and IP Assignment on Devices.

IP Address	CIDR Notation	Network Address	Broadcast Address	Usable Hosts
192.168.10.0	/25	192.168.10.0	192.168.10.127	192.168.10.1–192.168.10.126
192.168.10.128	/26	192.168.10.128	192.168.10.191	192.168.10.129–192.168.10.190
192.168.10.192	/27	192.168.10.192	192.168.10.223	192.168.10.193–192.168.10.222
192.168.10.224	/27	192.168.10.224	192.168.10.255	192.168.10.225–192.168.10.254
172.16.10.0	/30	172.16.10.0	172.16.10.3	172.16.10.1–172.16.10.2
192.169.1.0	/24	192.169.1.0	192.169.1.255	192.169.1.1–192.169.1.254
192.168.100.0	/24	192.168.100.0	192.168.100.255	192.168.100.1–192.168.100.254
10.1.1.0	/30	10.1.1.0	10.1.1.3	10.1.1.1–10.1.1.2
10.2.1.0	/30	10.2.1.0	10.2.1.3	10.2.1.1–10.2.1.2

Device	IP Address	Subnet Mask	DNS Address	Default Gateway
Branch-DHCP	192.168.10.226	255.255.255.224	172.16.10.2	192.168.10.225
DNS-Server	172.16.10.2	255.255.255.252	0.0.0.0	172.16.10.1
DHCP-Server	192.169.1.2	255.255.255.0	172.16.10.2	192.169.1.1
WEB-Server	192.168.100.100	255.255.255.0	172.16.10.2	192.168.100.5

VLAN ID	Default Gateway
10	192.168.10.1
20	192.168.10.129
30	192.168.10.193

- Branch Router

Building configuration...

Current configuration : 1441 bytes

!

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname Branch-Router
ip cef
no ipv6 cef
license udi pid CISCO2911/K9 sn FTX15245UKR-
no ip domain-lookup
spanning-tree mode pvst
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.128
ip helper-address 192.168.10.226
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.10.129 255.255.255.192
ip helper-address 192.168.10.226
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.10.193 255.255.255.224
ip helper-address 192.168.10.226
interface GigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.10.225 255.255.255.224
interface GigabitEthernet0/1
description HQ-Link
no ip address
duplex auto
speed auto
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
interface Serial0/3/0
ip address 10.1.1.2 255.255.255.252
!
```

```

interface Serial0/3/1
no ip address
clock rate 2000000
interface Vlan1
no ip address
router rip
version 2
network 10.0.0.0
network 192.168.10.0
no auto-summary
ip classless
ip flow-export version 9
banner motd ^CAuthorized Personal Only^C
line con 0
line aux 0
line vty 0 4
login
end

```

- Branch Switch

Building configuration...

```

Current configuration : 1709 bytes
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname Branch-SW
no ip domain-lookup
spanning-tree mode pvst
spanning-tree extend system-id
interface FastEthernet0/1
switchport access vlan 10
switchport mode access
interface FastEthernet0/2
switchport access vlan 10
switchport mode access
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
interface FastEthernet0/4
switchport access vlan 20
switchport mode access
interface FastEthernet0/5
switchport access vlan 20
switchport mode access

```

```

interface FastEthernet0/6
switchport access vlan 20
switchport mode access
interface FastEthernet0/7
switchport access vlan 30
switchport mode access
interface FastEthernet0/8
switchport access vlan 30
switchport mode access
interface FastEthernet0/9
switchport access vlan 30
switchport mode access
interface FastEthernet0/10
switchport access vlan 40
switchport mode access
interface GigabitEthernet0/1
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
interface GigabitEthernet0/2
interface Vlan1
no ip address
shutdown
banner motd ^CAuthorized Only^C
line con 0
line vty 0 4
login
line vty 5 15
login
end

```

- HQ-Router

Building configuration...

```

Current configuration : 1255 bytes
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname HQ-Router
ip cef

```

```
no ipv6 cef
license udi pid CISCO2911/K9 sn FTX15243DCK-
no ip domain-lookup
spanning-tree mode pvst
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
interface GigabitEthernet0/0.11
encapsulation dot1Q 11
ip address 172.16.10.1 255.255.255.252
interface GigabitEthernet0/0.12
encapsulation dot1Q 12
ip address 192.169.1.1 255.255.255.0
ip helper-address 192.169.1.2
interface GigabitEthernet0/1
description Branch-Link
no ip address
duplex auto
speed auto
interface GigabitEthernet0/2
description WEB-Link
no ip address
duplex auto
speed auto
interface Serial0/3/0
ip address 10.1.1.1 255.255.255.252
clock rate 64000
interface Serial0/3/1
description WEB-Link
ip address 10.2.1.1 255.255.255.252
interface Vlan1
no ip address
shutdown
router rip
version 2
```

```
network 10.0.0.0
network 172.16.0.0
network 192.169.1.0
no auto-summary
ip classless
ip flow-export version 9
banner motd ^CAuthorized Personal Only^C
line con 0
line aux 0
line vty 0 4
login
end
```

- HQ-Switch

Building configuration...

```
Current configuration : 1351 bytes
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname HQ-SW
no ip domain-lookup
spanning-tree mode pvst
spanning-tree extend system-id
interface FastEthernet0/1
switchport access vlan 11
switchport mode access
interface FastEthernet0/2
switchport access vlan 12
switchport mode access
interface FastEthernet0/3
switchport access vlan 12
switchport mode access
interface GigabitEthernet0/1
```

```
switchport trunk allowed vlan 11-12
switchport mode trunk
interface GigabitEthernet0/2
interface Vlan1
no ip address
shutdown
banner motd ^CAuthorized Personal Only^C
line con 0
line vty 0 4
login
line vty 5 15
login
end
```

- Web-Router

Building configuration...

Current configuration : 1023 bytes

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
hostname WEB-Router
no ip cef
no ipv6 cef
license udi pid CISCO2911/K9 sn FTX1524B1O4-
no ip domain-lookup
spanning-tree mode pvst
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
shutdown
interface GigabitEthernet0/1
```

```
ip address 192.168.100.5 255.255.255.0
duplex auto
speed auto
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
interface Serial0/3/0
description HQ-Link
ip address 10.2.1.2 255.255.255.252
clock rate 2000000
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
interface Vlan1
no ip address
shutdown
router rip
version 2
network 10.0.0.0
network 192.168.100.0
no auto-summary
ip classless
ip flow-export version 9
banner motd ^CAuthorized Personal Only^C
line con 0
line aux 0
line vty 0 4
login
end
```